The backfire effect in the perception and judgement of congruent and incongruent interpretation of political propaganda: A pre-analysis plan

ABSTRACT

This study investigates psychological barriers to the parsing and acceptance of opposing opinions or opinions that deviate from existing ones. Specifically, we focused on the Backfire Effect in the perception and judgement of information on Covid-related policies. Backfire effect refers to people's increased reliance on existing beliefs that the information presented to them was supposed to rectify. Previous research has focused on the causes and categorical stimulation of the effect. We hypothesize that the effect is a linear or qusai-linear function of the deviation from perceiver's existing belief. We manipulate the distance to the central belief of the effectiveness of "Dynamic-zero" policy and predict a longer distance will result in larger backfire effect. A perception and judgement task that includes 18 news abstracts reporting positive or negative consequences of "Dynamic-zero" policy in China was proposed. Future participants will be invited to read and rate the reliability of each of the abstracts, then a distractor task will empty participant's working memory. They will answer a questionnaire consisting of forced choices, rating, and open-ended questions. Pre-analysis plan, if properly executed, may reveal a non-categorical mental representation of information. Implications on public opinion management will be discussed.

OBJECTIVES

- **The Backfire Effect** (e.g., Lewandowsky, Ecker, Seifert, Schwarz, & Cook, 2012, Nythan & Reifler, 2010) states that an increased belief in what information presented to perceivers intends to rectify is often expected among people.
- Multiple types of backfire effect have been identified, including **worldview** and familiarity backfire. This study focuses on the former, where people's general belief of how the world and society should function is challenged.
- Previous studies have revealed Continued Influence Effect (e.g., Johnson & Seifert, 1994), statistical memory model (e.g., Hintzman & Curran, 1994), and Motivational Reasoning (e.g., Baumeister and Newman, 1994) as potential accounts for the backfire.
- However, much fewer studies have focused on **whether the effect is conditioned by the distances to existing beliefs, or if there are any fine-grained disagreements within the same opinion category.**
- Therefore, this study aims to investigate:
  - 1. Do people exhibit different degrees of backfire effect when presented with information that is congruent versus incongruent with their belief?
  - 2. For polarizing social issues, such as policies on Covid-19 containment, does fine-grained incongruency between people's opinion and presented information trigger backfire effect?
  - 3. And if it does, is the size of the effect relative to the degree of deviation from one's opinion?

METHODS

- A perception experiment was proposed, consisting of four phases (Figure 1).

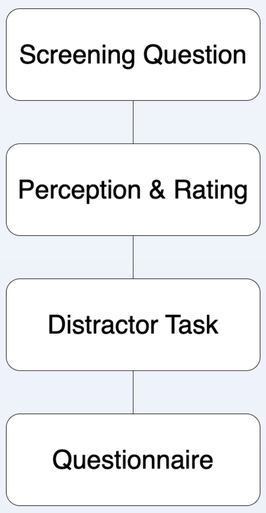

Figure 1

- **Screening question** explicitly asked for participant's stance on the "Dynamic-zero" policy: supportive or not.
- **Perception and rating task** exposed participants to positive/negative and individual/generic (See below) abstracts and asked for their rating of reliability. All abstracts presented in this phase were news digests from Chinese media.
- **The distractor task** emptied perceiver's working memory using speed mathematical calculations.
- **A post-exposure questionnaire** measured participants responses to similar Covid policies outside Mainland China.

- Grouping: Reference to news abstracts that were presented, simulating official responses to negative/positive consequences of "Dynamic-zero" policy (Table1).

Types of responses	Positive Events	Negative Events
Positive and individual-based responses to	Group 1	Group 2
Positive and generalized responses to	Group 3	Group 4

Table 1

- Generalized response: The scope of response is generalized outside the single event, such as claiming that the policy is successful in a larger region using effective control of infection in a village.

METHODS (Cont'd)

- Individual-based response: The scope of response is limited to the single events, such as attributing misconducts in mass covid-testing to the testing company, instead of systematic mis-supervision of CDC or pandemic prevention office.
- A complete list of stimuli and questionnaire items can be obtained from the author.
- The experiment is programmed using PsychoPy (2019) and was planned to be hosted on Pavlovia.org, an online experimenting platform.
- **Anonymity is guaranteed** by masking participant's IP address and reframing from collecting any personal identifiers. With all privacy protections in place, we expect participant's responses to be reliable.
- Due to rapid adjustments made by the Chinese Government, public opinion changed so fast that we were unable to conduct this experiment.

ANALYSIS & EXPECTED RESULTS

- We hypothesized that the backfire effect should be a linear function of the deviation from participant's existing beliefs, therefore, if assuming a neutral stance on "Dynamic-zero" policy, we expected the reliability rating to be ranked in an order of: **Group 1 > Group 2 > Group 3 > Group 4, inverse to the size of backfire effect.**
- For group 1, the quality and the scope of response was congruent with the events, stimulating less backfire effect and higher rating. For G2, the scope was congruent, but the quality was not. The quality was congruent in G3, but the scope is not. And for G4, both factors were incongruent.
- However, the ranking between G2 and G3 would depend on whether there would be a bias favoring quality or scope.
- Multiple analyses were planned to test our hypotheses statistically. Rating and responses in the questionnaire phase will be modelled using separate Linear Mixed-Effects Regressions. A planned formula is below, where quality, scope, two-way interaction between them, and stance as fixed effects, participant having random intercept, and item having by-subject random slopes, although a step-down would be implemented, and the actual regression may differ.
  - **Response ~ Quality * Scope + Stance + (1+Item|Participant)**
- We expect all fixed effects to be significant.

CONCLUSIONS & IMPLICATIONS

- If the results were to be the same as what we predicted, we may conclude:
  - **The Backfire Effect may happen in a continuous instead of purely categorical way.**
  - **The effect is influenced by distance to perceiver's existing social belief.**
  - **Quality (or scope, depending on statistical coefficients) of perceived information is weighted more heavily than the other factor.**
- The study may also have some implications on how information and beliefs in information are stored in mental reality:
  - **Instead of purely categorical representation of opposing views on a social issue, people can in fact detect more fine-grained differences between two pieces of information that share the same basic stance.**
  - **Such detailed differences may lead to a linear increase/decrease of backfire effect, or other psychological barriers to the dissemination of factual information.**

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